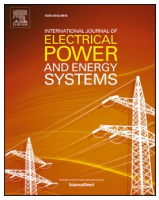
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Evaluation of the inertia distribution performance for multi-machine power system

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ABSTRACT

Given large-scale modern power systems with inverter-based resources, the frequency stability faces great challenges in the characteristics of the decreasing inertia for the system and the uneven distribution of the inertia. Although several metrics for system inertia performance have been developed, a more generalized inertia evaluation method is still required. This paper proposed an inertia evaluation method for a multi-machine power system considering both the inertia and disturbance distribution. To reflect the dynamic of a more general system, an equivalent homogeneous transformation (EHT) is used in the system frequency response model with heterogeneous parameters. To reveal the inertia distribution under different disturbances scenario, a general inertia (GI) is proposed to evaluate the nodal frequency stability. Finally, the effectiveness of the proposed equivalent homogeneous system and the metric is validated using two test systems: a modified 4-machine 2-area power system and the 5-area 68-bus system. Numerical simulation study reveals that the proposed method can effectively reflect the inertia spatial distribution characteristics.

1. Introduction

The aggressive vision of carbon peak and carbon neutrality as well as the rapid and significant growth of renewable power in China over the past few years has attracted great attention around the world. However, most of the renewable energy are inverter-based resources (IBRs), including PV, wind power, battery storage, etc. Different from the synchronous generators (SG), usually IBRs do not have natural physical inertia support to power systems, which will definitely reduce the overall system equivalent inertia [1,2]. Thus, the decreased system inertia has resulted in an unacceptable levels of rate of change of frequency (ROCOF) and result in a surge of frequency-related incidents has been observed, including the inadvertent triggering of over-/under-frequency relays, load-shedding, and special protection schemes [3,4]. These were evidenced by incidents that occurred in Australia in 2016 and the United Kingdom in 2019, which exposed the role of inertia in frequency stability [5–7]. To address these issues caused by low inertia, an accurate estimation of inertia is needed. According to [8], the inertia estimation methodologies are categorized as measurement-based and model-based approaches.

As the advancement of phasor measurement units (PMUs) in the wide-area measurement systems, measurement-based approaches have been undertaken by both industry and academia to identify and measure varying degrees of inertia within power systems through the utilization of real-time monitoring [9–11]. The application of curve fitting methodology is proposed for the estimation of frequency response (FR), while the acquisition of inertia is achieved through the utilization of ROCOF estimation within a specified time interval [11,12]. To improve the accuracy of frequency signals and inertia estimation, the authors in [10] use a low-pass filter for noise mitigation, which introduces a time delay. Further, a Bayesian framework is proposed in [13] that applies PMU data to accurately estimate generator inertia. However, due to its high computational complexity, the practical implementation of this method for online monitoring is limited. In short, the measurement-based methods are employed in dynamic power system operations and may not be suitable for power system planning which heavily relies on the power system models.

Therefore, many works have been conducted on model-based inertia estimation approaches. In [14], the time-domain simulation (TDS) technique is applied to investigate the effects of inertia spatial distribution on power system dynamics, followed by modal analysis utilizing the variable projection method (VPM) to quantitatively assess the influence of inertia location. The authors in [15] propose an TDS-based index to quantify the electrical distance from any bus to the center of

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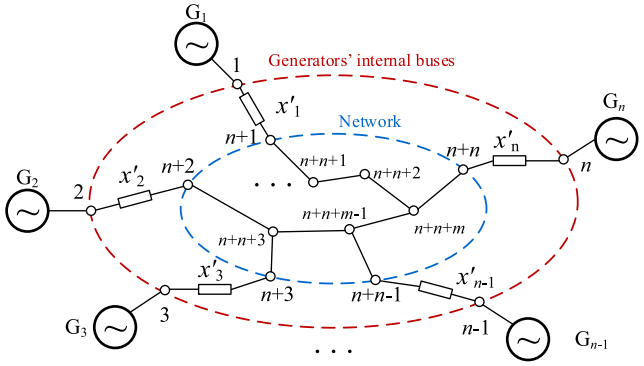


Fig. 1. Topology of the multi-machine system.

inertia (COI) and higher index value can reflect a relatively larger distance. The index is further employed for actuator placement to enhance grid dynamic performance [16]. However, these TDS-based approaches are disturbance dependent and may give rise to considerable computational burden for large-scale system application.

On the other, some analytical methods are developed. In [17,18], a linearized equivalent model considering the interdependence between active power and phase angle is developed to represent an aggregated rotor dynamics during frequency response (FR), which is called the average system-frequency (ASF) model. Based on ASF, the authors in [19] further aggregate the dynamic of primary frequency control loop in a multi-machine power system, but it can only evaluate the dynamic of the COI and ignores the spatial–temporal characteristics of inertia distribution [20,21]. In [22], the bus frequency is decomposed into a COI component and other frequency deviation components to identify the differences in inertia response between nodes. The decomposition results show that the traditional frequency stability analysis method mainly based on the COI response may neglect the transient components of inter-area oscillations. Therefore, it is urgent to develop techniques for detecting and measuring the inertia distribution in modern power systems, which can improve system operation and short-term planning like optimal generation scheduling [23,24] and inertia allocation [25,26]. In this regard, some pioneering works have conducted on the inertia distribution by means of network topology analysis and inertial source placement. [27] defines a nodal inertia concept to represent the impact of frequency deviation components on frequency dynamics for network buses, which is based on the frequency component-decoupling theory in [22]. However, in [27] there is a strict assumption of generator dynamic parameter proportionality, commonly referred to as parameter homogeneity. In [28], an analytical formulation considering generator location is proposed and the unit step disturbance is imposed on the generator internal buses to validate their inertia distribution estimation. One of the major problems is that it neglects the influence of disturbance propagation through the system. Thus, [29] proposes another analytical expression considering disturbance location, but it is not suitable for multi-disturbance scenarios.

Based on the above analysis, this paper proposes an estimation method to evaluate inertia distribution considering the multi-machine model considering heterogeneous parameters and multi-disturbance scenarios. The contribution of this paper can be summarized as follows:

1) An equivalent FR model is proposed, wherein the dynamic parameters of generators are transformed to meet parameter homogeneous conditions, so that the model can reflect the frequency dynamics of the system with heterogeneous parameters.
2) An analytical formulation based on the constructed equivalent FR model is introduced to quantify the nodal inertia, which can identify the buses with lower inertia under different power disturbances and serve as a tool for further analysis to improve nodal frequency stability.

The paper is organized as follows: In Section 2, a multi-machine FR model is firstly analytically derived; In Section 3, an equivalent homogeneous transformation is then developed to consider heterogeneous parameters in the system; In Section 4, a general inertia (GI) metric is proposed and a GI evaluation procedure is presented; Section 5 provides case studies to validate the proposed method, and finally conclusions are drawn in Section 6.

2. Multi-machine frequency dynamics modelling considering grid topology

2.1. Network modelling

Considering a multi-machine electrical network as illustrated in Fig. 1: $\phi_G = \{1,...,n\}$,$\phi_{NI} = \{n+1,...,n+n+m\}$ denote the index sets of the generator internal buses and non-internal buses (including buses of the generator terminals and loads), where there are n generators and m loads. Then, by linearizing the power flow equations with the power and angle elements retained:

$$\Delta\mathbf{P}_e = \begin{bmatrix} \mathbf{L}_{11} & \mathbf{L}_{12} \\ \mathbf{L}_{21} & \mathbf{L}_{22} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\theta}_G \\ \Delta\boldsymbol{\theta}_{NI} \end{bmatrix} \tag{1}$$

where $\Delta\mathbf{P}_e$, $\Delta\boldsymbol{\theta}_G$ and $\Delta\boldsymbol{\theta}_{NI}$ denote the vectors of the power, generator internal and non-internal buses variations, respectively; $\mathbf{L}_{11} \in \mathbb{R}^{n\times n}$, $\mathbf{L}_{12} \in \mathbb{R}^{n\times(n+m)}$, $\mathbf{L}_{21} \in \mathbb{R}^{(n+m)\times n}$, $\mathbf{L}_{22} \in \mathbb{R}^{(n+m)\times(n+m)}$ are the block matrixes of the Laplacian matrix $\mathbf{L}$ and the entries in $\mathbf{L}$ are

$$\begin{cases} L_{ij} = -V_i^{(0)} V_j^{(0)} B_{ij} cos\left(\theta_i^{(0)} - \theta_j^{(0)}\right) \\ L_{ii} = V_i^{(0)} \sum_{j=1, j\neq i}^{n+n+m} V_j^{(0)} B_{ij} cos\left(\theta_i^{(0)} - \theta_j^{(0)}\right) \end{cases} \tag{2}$$

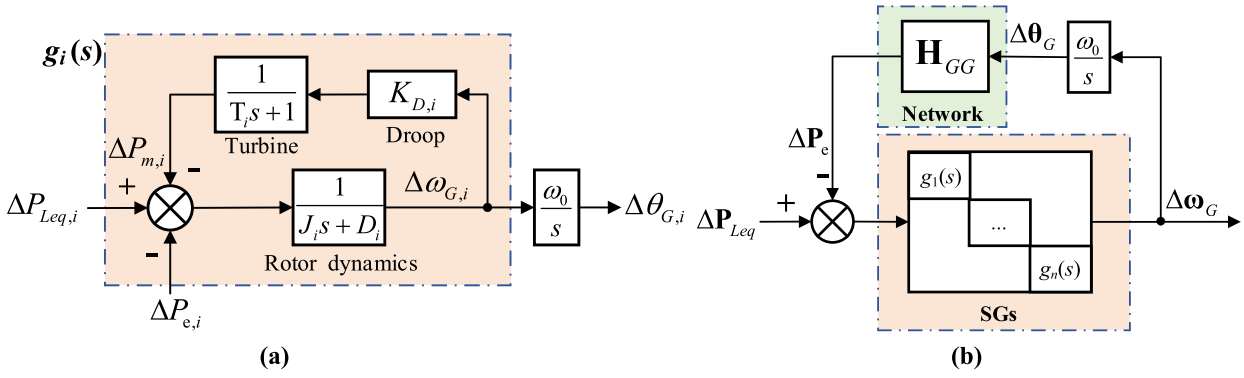


Fig. 2. Frequency response of multi-machines (a) Single machine frequency response model (b) Multi-machine frequency response model.

where L_{ij} and L_{ii} are the off-diagonal and diagonal elements of $\mathbf{L}$; B_{ij} is the element in the susceptance matrix; $V_i^{(0)}$, $\theta_i^{(0)}$ are the steady-state value of voltage magnitude and angle of the i-th bus.

2.2. Frequency response model

A typical single machine frequency response model $g_i^{-1}(s)$ ($i = 1, 2, \ldots, n$) can be found in [30] and illustrated in Fig. 2(a). Further, the frequency response of a multi-machine system can be written as:

$$\begin{bmatrix} \omega_0^{-1}\mathbf{g}^{-1}(s)s\Delta\boldsymbol{\theta}_G \\ \mathbf{0} \end{bmatrix} = -\begin{bmatrix} \mathbf{L}_{11} & \mathbf{L}_{12} \\ \mathbf{L}_{21} & \mathbf{L}_{22} \end{bmatrix}\begin{bmatrix} \Delta\boldsymbol{\theta}_G \\ \Delta\boldsymbol{\theta}_{NI} \end{bmatrix} + \begin{bmatrix} \Delta\mathbf{P}_{LG} \\ \Delta\mathbf{P}_{LB} \end{bmatrix} \tag{3}$$

$$\mathbf{g}^{-1}(s) = diag\left(g_i^{-1}(s)\right) \tag{4}$$

with $g_i(s) = \frac{T_i s+1}{J_i T_i s^2+(D_i T_i+J_i)s+D_i+K_{D,i}}$

where $\Delta\mathbf{P}_{LG} = [\Delta P_1, \ldots, \Delta P_n]^{\mathrm{T}}$ and $\Delta\mathbf{P}_{LB} = [\Delta P_1, \ldots, \Delta P_k, \ldots, \Delta P_{n+n+m}]^{\mathrm{T}}$ ($k \in \phi_{NI}$) represent the injected power variation at the internal and non-internal buses, respectively. For the purpose of frequency response analysis, it is assumed that there is no instant injected power variation at internal buses when load step suddenly happens, i.e., $\Delta\mathbf{P}_{LG} = \mathbf{0}$. It should be noted that $\Delta\mathbf{P}_{LB}$ can be used to represent not only the load variation, but also including power output variation at those generator terminals.

According to Appendix A, the frequency response for the multi-machine system can be derived as follows:

$$\Delta\boldsymbol{\omega}_G = \mathbf{J}^{-1/2}\left(\tilde{\mathbf{g}}^{-1}(s) + \omega_0\mathbf{H}_J/s\right)^{-1}\mathbf{J}^{-1/2}\Delta\mathbf{P}_{Leq} \tag{5}$$

$$\begin{cases} \mathbf{H}_J = \mathbf{J}^{-1/2}\mathbf{H}_{GG}\mathbf{J}^{-1/2} \\ \mathbf{J} = diag(J_i) \\ \mathbf{H}_{GG} = \mathbf{L}_{11} - \mathbf{L}_{12}\mathbf{L}_{22}^{-1}\mathbf{L}_{21} \\ \Delta\mathbf{P}_{Leq} = -\mathbf{L}_{12}\mathbf{L}_{22}^{-1}\Delta\mathbf{P}_{LB} \end{cases} \tag{6}$$

$$\begin{cases} \tilde{\mathbf{g}}^{-1}(s) = diag\left(\tilde{g}_i^{-1}(s)\right) \\ \tilde{g}_i^{-1}(s) = g_i^{-1}(s)/J_i = s + \left(D_i + \dfrac{K_{D,i}}{T_i s+1}\right)\Big/ J_i \end{cases} \tag{7}$$

where $\Delta\boldsymbol{\omega}_G$ is the frequency variation of the generator internal buses; $\mathbf{H}_J$ is the weighted susceptance matrix; $\mathbf{H}_{GG}$ is the Kron-reduced matrix; $\mathbf{J}$ is the associated diagonal matrix for a one-dimension array of inertia constant of generators; $\Delta\mathbf{P}_{Leq}$ is the vector of equivalent electromagnetic power variation at internal buses induced by load disturbance; $\tilde{\mathbf{g}}^{-1}(s)$ is diagonal matrix of weighted transfer functions of generators; D_i is the damping factor of generators while $K_{D,i}$, and T_i are the droop coefficient and the time constant of the turbine control, respectively.

3. Equivalent homogeneous model

3.1. Homogeneous-condition based model

When the following parameter homogeneity conditions are satisfied, the system model is called a homogeneous-condition based model [22]:

$$\begin{cases} T_i = T \\ J_i : D_i : K_{D,i} = J_{sum} : D_{sum} : K_{D,sum} \end{cases} \tag{8}$$

where $J_{sum} = \sum J_i$, $D_{sum} = \sum D_i$ and $K_{D,sum} = \sum K_{D,i}$, then (5) can be rewritten as:

$$\begin{aligned} \Delta\boldsymbol{\omega}_G &= \mathbf{J}^{-\frac{1}{2}}\left(g_0^{-1}(s)\mathbf{I}_n + \omega_0\mathbf{H}_J/s\right)^{-1}\mathbf{J}^{-\frac{1}{2}}\Delta\mathbf{P}_{Leq} \\ &= \mathbf{J}^{-\frac{1}{2}}\mathbf{U}\left[g_0^{-1}(s)\mathbf{I}_n + \omega_0\boldsymbol{\Lambda}/s\right]^{-1}\mathbf{U}^{\mathrm{T}}\mathbf{J}^{-\frac{1}{2}}\Delta\mathbf{P}_{Leq} \\ &= \mathbf{J}^{-\frac{1}{2}}\mathbf{U}\mathrm{diag}(G_l(s))\mathbf{U}^{\mathrm{T}}\mathbf{J}^{-\frac{1}{2}}\Delta\mathbf{P}_{Leq} \end{aligned} \tag{9}$$

$$g_0^{-1}(s) = s + \left(D_{sum} + \frac{K_{D,sum}}{Ts+1}\right)\Big/ J_{sum} \tag{10}$$

$$G_l(s) = \frac{1}{g_0^{-1}(s) + \omega_0\lambda_l/s} \tag{11}$$

where $\boldsymbol{\Lambda}$ and $\mathbf{U}$ are the eigenvalue and eigenvector matrices of $\mathbf{H}_J$, respectively, i.e., $\mathbf{H}_J = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathrm{T}}$; $\boldsymbol{\Lambda} = \mathrm{diag}\{\lambda_l\}$ ($l = 1, 2, \ldots, n$), and λ_l is the l-th eigenvalue of $\mathbf{H}_J$; and $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n]$ is an orthogonal matrix, i.e., $\mathbf{U}\mathbf{U}^{\mathrm{T}} = \mathbf{I}_n$. Let $\mathbf{A}_l = \mathbf{J}^{-1/2}\mathbf{u}_l$, and the frequency responses described in (9) can be further rewritten as:

$$\begin{aligned} \Delta\boldsymbol{\omega}_G &= \mathbf{J}^{-1/2}\mathbf{U}\mathrm{diag}(G_l(s))\mathbf{U}^{\mathrm{T}}\mathbf{J}^{-1/2}\Delta\mathbf{P}_{Leq} \\ &= \mathbf{A}_1\mathbf{A}_1^{\mathrm{T}}G_1(s)\Delta\mathbf{P}_{Leq} + \sum_{l\geq 2}^{n}\mathbf{A}_l\mathbf{A}_l^{\mathrm{T}}G_l(s)\Delta\mathbf{P}_{Leq} \\ &= \Delta\boldsymbol{\omega}_{G(1)}(s) + \sum_{l\geq 2}^{n}\Delta\boldsymbol{\omega}_{G(l)}(s) \end{aligned} \tag{12}$$

where $\Delta\boldsymbol{\omega}_{G(l)}(s) = \mathbf{A}_l\mathbf{A}_l^{\mathrm{T}}G_l(s)\Delta\mathbf{P}_{Leq}$ ($l = 1, 2, \ldots, n$) is the particularly introduced frequency component related to the l-th FR dynamic G_l(s).

Further, as for the (n−1)-rank semidefinite matrix $\mathbf{H}_J$, it always satisfies that the first eigenvalue $\lambda_1 = 0$ and the first eigenvector $\mathbf{u}_1 = J_{sum}^{-1/2}\mathbf{J}^{1/2}\mathbf{1}_{n\times 1}$ after the eigenvalue decomposition [22]. Hence, $\Delta\omega_{G(1)}$ can be derived as:

$$\begin{aligned} &\Delta\boldsymbol{\omega}_{G(1)}(s) = \mathbf{A}_1\mathbf{A}_1^{\mathrm{T}}G_1(s)\Delta\mathbf{P}_{Leq} \\ &= \frac{Ts+1}{J_{sum}Ts^2 + (TD_{sum} + J_{sum})s + K_{D,sum} + D_{sum}}\mathbf{1}_{\mathrm{n}\times\mathrm{n}}\Delta\mathbf{P}_{Leq} \end{aligned} \tag{13}$$

In (12)-(13), frequency responses can be decomposed into a group of frequency components, which can be roughly classified into two categories, i.e., the system frequency term $\Delta\omega_{G(1)}$, and the transient terms $\Delta\omega_{G(l)}$ ($l > 1$). It should be noted that as all the entries of $\mathbf{A}_1\mathbf{A}_1^{\mathrm{T}}$ is $1/J_{sum}$, so the term $\Delta\omega_{G(1)}$ is related to the total system dynamic parameters and independent of network topology. On the other hand, the steady-state value of the transient terms $\Delta\omega_{G(l)}$ is zero and thus called the transient terms here.

3.2. Equivalent homogeneous transformation (EHT)

In practical power systems, most generators have distinct frequency regulation models, i.e., there exists model heterogeneity issues including: 1) Structure heterogeneity, which can be solved by some identical structure transformation methods in [31]; and 2) Parameter heterogeneity. Since the derivation in last subsection is based on the parameter homogeneous conditions, which cannot be satisfied in most multi-machine systems, an EHT is developed to solve the issue.

According to the derivation in last subsection, a general system usually cannot be decomposed into different frequency components. An EHT can be developed as follows:

$$\overline{J}_i = J_i,\ \overline{D}_i = \frac{J_i}{J_{sum}}D_{sum},\ \overline{K}_{D,i} = \frac{J_i}{J_{sum}}K_{D,sum} \tag{14}$$

$$\overline{T}_i = \sum_{i=1}^{n} J_i T_i / J_{sum} = \overline{T} \tag{15}$$

where the variables with overbar denote the post-transformation parameters. From (14), it can be easily obtained:

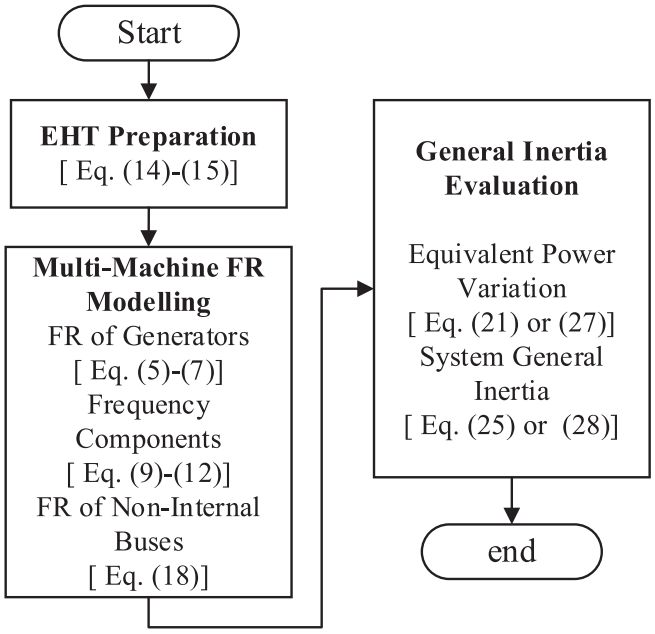


Fig. 3. Evaluation of inertia distribution performance.

$$\overline{J}_i : \overline{D}_i : \overline{K}_{D,i} = J_{sum} : D_{sum} : K_{D,sum} \tag{16}$$

After the transformation (14)-(15), the following equivalent homogeneous item $\overline{g}_0^{-1}(s)$ can be obtained to replace $g_0^{-1}(s)$ in (9):

$$\overline{g}_0^{-1}(s) = s + \left(D_{sum} + \frac{K_{D,sum}}{\overline{T}s+1}\right) \Big/ J_{sum} \tag{17}$$

4. Evaluation of inertia distribution performance

In this section, the general inertia (GI) concept will be developed to evaluate the inertia distribution performance based on the post-EHT model.

4.1. General inertia J_{GI}

To further assess the inertia performance at all buses, the relationship between the frequency of internal and non-internal buses needs to be developed. From A.ppendix A, we have:

$$\begin{aligned} \Delta\boldsymbol{\omega}_{\mathrm{B}} &= -\mathbf{L}_{22}^{-1}\mathbf{L}_{21}\Delta\boldsymbol{\omega}_{\mathrm{G}} = \mathbf{D}\Delta\boldsymbol{\omega}_{\mathrm{G}} \\ &= \mathbf{D}\sum\nolimits_{l=1}^{\mathrm{n}} \mathbf{A}_l\mathbf{A}_l^{\mathrm{T}}\mathrm{G}_l(s)\Delta\mathbf{P}_{\mathrm{Leq}} \end{aligned} \tag{18}$$

where $\Delta\boldsymbol{\omega}_B$ is the frequency vector of non-internal buses; $\mathbf{D}$ is similar to the frequency divider formulation [32] and $\mathbf{D} = -\mathbf{L}_{22}^{-1}\mathbf{L}_{21} \in \mathbb{R}^{(n+m)\times n}$. Then, $\Delta\mathbf{P}_{Leq}$ in (6) can be rewritten as

$$\Delta\mathbf{P}_{Leq} = -\mathbf{L}_{12}\mathbf{L}_{22}^{-1}\Delta\mathbf{P}_{LB} = \mathbf{D}^{\mathrm{T}}\Delta\mathbf{P}_{LB} \tag{19}$$

To ease explanation by preserving the original bus indices, a new $(2n + m)\times(2n + m)$ matrix $\mathbf{C}$ is constructed as follows

$$\mathbf{C} = \{c_{ij}\} = \begin{bmatrix} \mathbf{0}_{n\times n} & \mathbf{0}_{n\times(n+m)} \\ \mathbf{D} & \mathbf{0}_{(n+m)\times(n+m)} \end{bmatrix} \tag{20}$$

Based on (19), the i-th entry of $\Delta\mathbf{P}_{Leq}$ at internal bus can be written as follows:

$$\begin{cases} \Delta P_{\mathrm{Leq},i} = \sum\limits_{k\in\phi_{NI}} c_{ki}\Delta P_k, \ i \in \phi_G \\ \Delta\mathbf{P}_{Leq} = \left[\Delta P_{Leq,1}, \ldots, \Delta P_{Leq,i}, \ldots, \Delta P_{Leq,n}\right]^{\mathrm{T}}/s \end{cases} \tag{21}$$

As the sum of each row in $\mathbf{D}$ matrix is 1, i.e., $\sum D_{ij} = 1$ [33], the following equation can be obtained by multiplying both sides of (19) with the row vector of unit entries,

$$\begin{aligned} \Delta P_{sum} &= \sum_{i\in\phi_G}\Delta P_{Leq,i} = \mathbf{1}_{n\times 1}^{\mathrm{T}}\Delta\mathbf{P}_{Leq} = \mathbf{1}_{n\times 1}^{\mathrm{T}}\mathbf{D}^{\mathrm{T}}\Delta\mathbf{P}_{LB} \\ &= \mathbf{1}_{(n+m)\times 1}^{\mathrm{T}}\Delta\mathbf{P}_{LB} = \sum_{k\in\phi_{NI}}\Delta P_k \end{aligned} \tag{22}$$

where ΔP_{sum} is the total power variation.

Given the initial value theorem, the relationship between the initial RoCoF at the non-internal buses $\mathbf{F}_B$ and the internal buses $\mathbf{F}_G$ is as follows:

$$\begin{cases} \mathbf{F}_B = \lim\limits_{s\to\infty} s^2\Delta\boldsymbol{\omega}_B = \mathbf{D}\lim\limits_{s\to\infty} s^2\Delta\boldsymbol{\omega}_G = \boldsymbol{D}\cdot\mathbf{F}_G \\ \mathbf{F}_G = \mathbf{A}\mathbf{A}^T\Delta\mathbf{P}_{Leq} = \mathbf{J}^{-\frac{1}{2}}\mathbf{U}\mathbf{U}^{\mathrm{T}}\mathbf{J}^{-\frac{1}{2}}\Delta\mathbf{P}_{Leq} = \mathbf{J}^{-1}\Delta\mathbf{P}_{Leq} \end{cases} \tag{23}$$

According to (23), the initial RoCoF at the k-th non-internal bus $F_{B,k}$ can also be given as:

$$F_{B,k} = \sum_{i\in\phi_G} c_{ki}F_{G,i} = \sum_{i\in\phi_G} c_{ki}\Delta P_{Leq,i} \Big/ J_i \ , \ k \in \phi_{NI} \tag{24}$$

where $F_{G,i}$ is the initial RoCoF at the i-th generator internal bus. Then, the general inertia (GI) of the k-th non-internal bus $J_{GI,k}$ can be derived as:

$$J_{GI,k} = \frac{\Delta P_{sum}}{F_{B,k}} = \Delta P_{sum} \Big/ \left(\sum_{i\in\phi_G}\frac{c_{ki}}{J_i}\Delta P_{Leq,i}\right), \ k \in \phi_{NI} \tag{25}$$

The analytical formulation $J_{GI,k}$ considers the power changes in different locations, which can be used to determine the buses mostly affected by the disturbances.

In particular, when there are uniform load changes happened at all non-internal buses, i.e., the variation is proportional to its initial or rated load,

$$\Delta P_k = \alpha\% P_k^{(0)}, \ k \in \phi_{NI} \tag{26}$$

where α% is power variation percentage, $P_k^{(0)}$ is the rated load at the k-th non-internal bus. Accordingly, the power disturbances induced at all internal buses can be easily derived from (21):

$$\begin{cases} \Delta P_{Leq,i} = \alpha\%\sum\limits_{k\in\phi_{NI}} c_{ki}P_k^{(0)} = \alpha\%P_{Leq,i}^{(0)} \\ \Delta\mathbf{P}_{Leq} = \alpha\%\left[P_{Leq,1}^{(0)}, \ldots, P_{Leq,i}^{(0)}, \ldots, P_{Leq,n}^{(0)}\right]^{\mathrm{T}}/s \end{cases} \tag{27}$$

where $P_{Leq,i}^{(0)} = \sum_{k\in\phi_{NI}} c_{ki}P_k^{(0)}$ denotes the initial equivalent injected power at the i-th internal bus. Then the proposed metric (25) has a simplified form:

$$J_{GI,k} = P_{sum}^{(0)} \Big/ \left(\sum_{i\in\phi_G}\frac{c_{ki}}{J_i}P_{Leq,i}^{(0)}\right), \ k \in \phi_{NI} \tag{28}$$

where $P_{sum}^{(0)} = \sum_{i\in\phi_G} P_{Leq,i}^{(0)} = \sum_{k\in\phi_{NI}} P_k^{(0)}$ is the total initial load.

Now let us discuss a more particular case of (27) in [28], given the following strong assumption:

$$\Delta P_{Leq,1} = \cdots = \Delta P_{Leq,i} = \cdots = \Delta P_{Leq,n}, \ i \in \phi_G \tag{29}$$

Then (25) can be simplified as follows [28]:

$$J_{GI,k} = n \Big/ \sum_{i\in\phi_G}\frac{c_{ki}}{J_i} \tag{30}$$

Eq. (28) shows that the GI is related to the steady load distribution and is a constant value under uniform disturbance with different α%. On the other hand, it is very interesting to notice that the metric (30) is a special form of (28) if all initial equivalent power in (27) are identical,

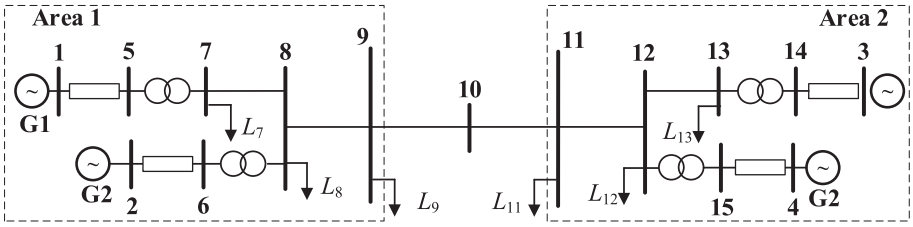


Fig. 4. The 4-machine 2-area single-line power system.

Table 1
SG fundamental parameters (p.u.).

SG	x'	X	r	T'_d	J	D
G1	0.0015	0.0136	0.001	5	10	2
G2	0.0015	0.0136	0.001	5	6	3
G3	0.0015	0.0136	0.001	5	8	4
G4	0.0015	0.0136	0.001	5	12	1

Table 2
SG primary frequency control parameters (p.u.).

SG	Droop K_D	T
G1	12	5
G2	13	6.5
G3	17	3.5
G4	22	4

Table 3
Parameters of the 4-machine 2-area power system.

Parameter		Value (per-unit values)
Generation Profile		$P_{ref1} = P_{ref2} = P_{ref3} = P_{ref4} = 0.75$
Load Profiles	Case 1	$L_7 = 0.3$, $L_8 = 0.3$, $L_9 = 0.3$, $L_{11} = 0.7$, $L_{12} = 0.7$, $L_{13} = 0.7$
	Case 2	$L_7 = 0.7$, $L_8 = 0.7$, $L_9 = 0.7$, $L_{11} = 0.3$, $L_{12} = 0.3$, $L_{13} = 0.3$

Table 4
Post-EHT parameters of SGs (p.u.).

SG	J	$\overline{D}$	$\overline{K}_D$	$\overline{T}$
1	10	2.5	18.9	4.125
2	6	1.5	11.3	4.125
3	8	2.0	15.1	4.125
4	12	3.0	22.7	4.125

which cannot be applied to those different power variation scenarios.

4.2. Evaluation of inertia distribution performance

According to the characteristics of the above metrics, an evaluation method is proposed to facilitate the analysis of the inertia distribution performance in the system, which is shown in Fig. 3.

Step 1 EHT Preparation: Transform heterogeneous parameters of generation units to meet the equivalent parameter homogeneous form with (14)-(15).

Step 2 Multi-machine FR Modelling: Obtain the FR model of generators (5)-(7). Get the frequency components (9)-(12) according to eigenvalue decomposition of $\mathbf{H}_J$. Calculate the frequency divider matrix $\mathbf{D}$ and further get frequency response model of non-internal buses (18).

Step 3 General Inertia Evaluation: Get the equivalent power variation using (21), or (27) for uniform disturbance. Evaluate the system general inertia using (25) or (28) for uniform disturbance.

5. Case study

To validate the effectiveness of the equivalent homogeneous model and the proposed inertia distribution metrics, the simulation based on MATLAB/Simulink is carried out in two test systems: (1) a modified 4-machine 2-area system and (2) the 5-area 68-bus system.

5.1. Modified 4-machine 2-area system

The modified 4-machine 2-area system illustrated in Fig. 4 considers generator internal buses. The rated voltage U_N is 230 kV and the rated capacity S_N is 900MVA. The SG frequency dynamics modelling is based on the fifth-order SG model [34], which is explained in A.ppendix B; the first-order primary frequency regulation model[22] illustrated in Fig. 2 (a) is used; the dynamic parameters of the generators and the primary frequency control are given in Table 1 and Table 2. The main network data can be found in [30] and the initial generation and load data are shown in Table 3, in which the two area load profiles are investigated.

5.1.1. Effectiveness of the equivalent homogeneous model

The post-EHT parameters for SGs are calculated according to (14)-(15), and the results are presented in Table 4.

To investigate the equivalence effectiveness, at t = 5 s a load step $\Delta P_9 = 0.1$p.u. at bus 9 is imposed. In Fig. 5(a)-(b), the time domain simulation (TDS) results are compared between the original parameters and the equivalent homogeneous parameters. It can be seen that the frequency responses between the original model and the post-EHT model are well consistent. Furthermore, the time-domain metrics of buses, including ROCOF and frequency nadir, are compared between the pre-EHT model and the post-EHT model in Fig. 5(c). The ROCOF remains consistent across all buses and the largest relative frequency errors of the post-EHT model with respect to the original model is less than 10 %, which is acceptable and hence verifies the accuracy of the equivalent model.

5.1.2. Effectiveness of the proposed metric

A comparison study is conducted between three inertia evaluation methods: 1) The method using Eq.(30) in [28]; 2) The proposed GI-based method; and 3) The TDS evaluation. It should be noted that the RoCoF based on the TDS evaluation are calculated in the first 0.15 s after the load step in order to reduce the noise impact.

1) Scenario 1: Uniform Disturbance

An uniform step power variation is set at all load buses in Area 1 and Area 2, changing α% in (26) from −15 % to 15 % with a step size 3 %. The inertia values are evaluated based on the procedure in last section and the inertia trajectory results of the three methods shown in Fig. 6, in which two cases corresponding to the two load levels in Table 3 are considered and Bus 7 (in area 1) and Bus 13 (in area 2) are chosen as the observation buses.

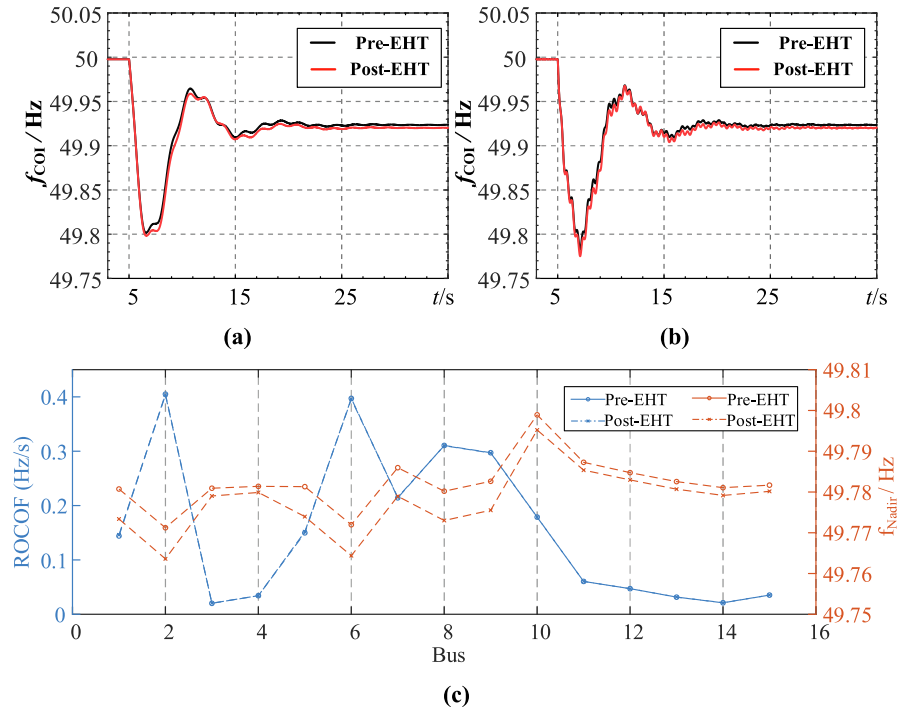


Fig. 5. Frequency dynamic comparison (a) Frequency of COI (b) Frequency of Bus 9 (c) Time-domain metrics of buses.

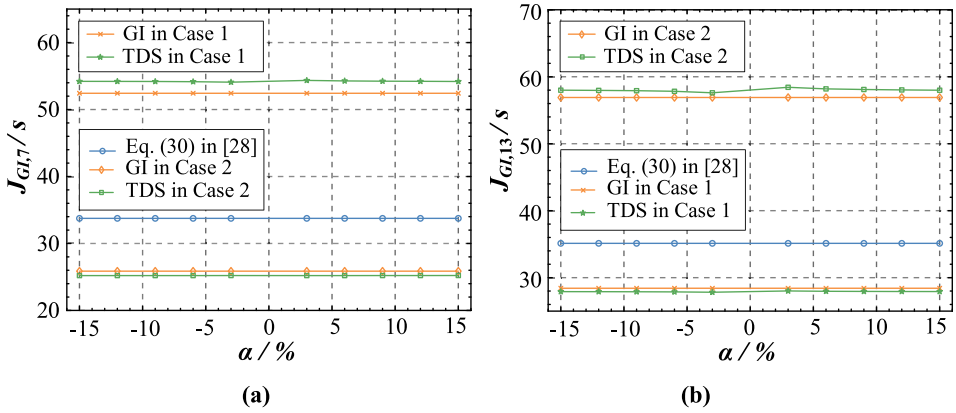


Fig. 6. Inertial value trajectory under uniform disturbance (a) Bus 7 (b) Bus 13.

In Fig. 6, it is very clear that the nodal inertia values calculated according to Eq. (30) in [28] are identical in the two cases, while the proposed metric GI has different values and can reflect inertial changes under multiple operation conditions, as discusses in IV-A. In addition, the inertia trajectory from the proposed GI-based method is more approximate to the TDS-based result, i.e., GI metric has better accuracy compared with [28]. Further, the inertia comparisons of buses with α%= 15 % is shown in Fig. 7, in which the similar observations can also be found.

The spatial distribution of the inertia is plotted in heatmap Fig. 8. In Case 1, Bus 1 and 5 exhibit the lowest nodal inertia values ($J_{GI,1} = 69.26$ s and $J_{GI,5} = 67.71$ s), whereas in case 2, Bus 4 and 15 demonstrate the lowest inertia (65.22 s and 64.36 s, respectively), which corresponds to two different operating conditions.

2) Scenario 2: Non-uniform Disturbance

It should be noted that the proposed GI metric (25) is suitable for any power variation distribution. Nevertheless, to make the conclusion more intuitive, the case 1 load profile is used here. The non-uniform load step here is set as follows: 1) a uniform load change with α%=3% imposed at all load buses; 2) simultaneously, an additional load variation imposed at L_7 and L_9 is added with amount changing from 2 % to 20 % of their rated value, with a step size 2 %. The inertia curves at Bus 7 and Bus 13 when loads change are shown in Fig. 9.

In Fig. 9, when the power changes are not uniform, the GI values are no longer constant and vary with disturbance. However, the inertia trends of GI and the TDS result is still consistent. The inertia tendency of the two buses in Bus 7 and Bus 13 are opposite in the presence of the increasing load step.

5.2. The 5-area 68-bus system

To verify the effectiveness and applicability of the proposed model and inertia evaluation method, the 5-area 68-bus system in [35] is further modified with 4 grid-forming IBRs with the virtual inertia control installed at Bus 3, Bus 6, Bus 9 and Bus 12 to replace the original SGs, as illustrated in Fig. 10. There are five areas in the system with 12

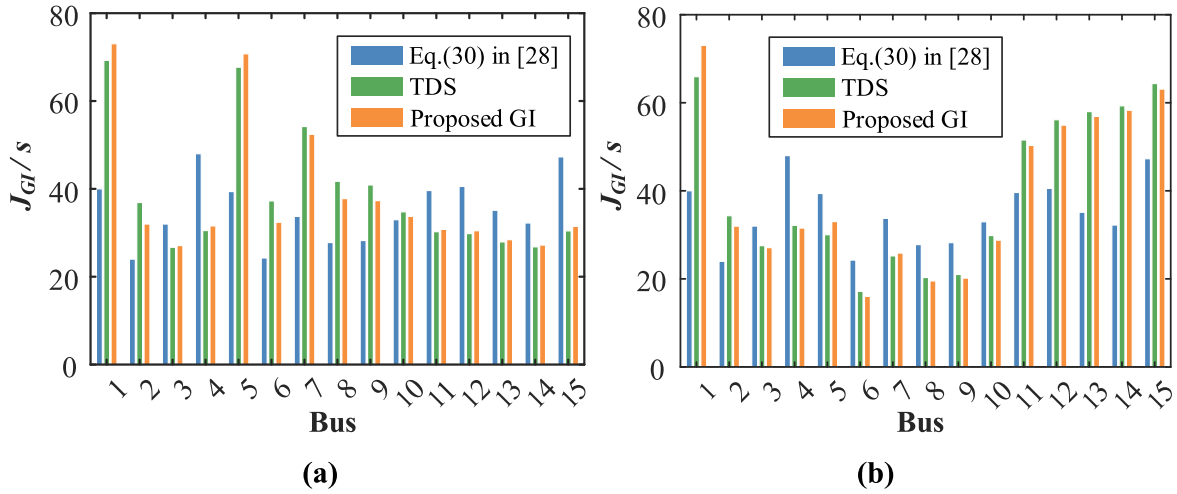


Fig. 7. Inertia comparisons of different methods (α%=15 %) (a) Case 1 (b) Case 2.

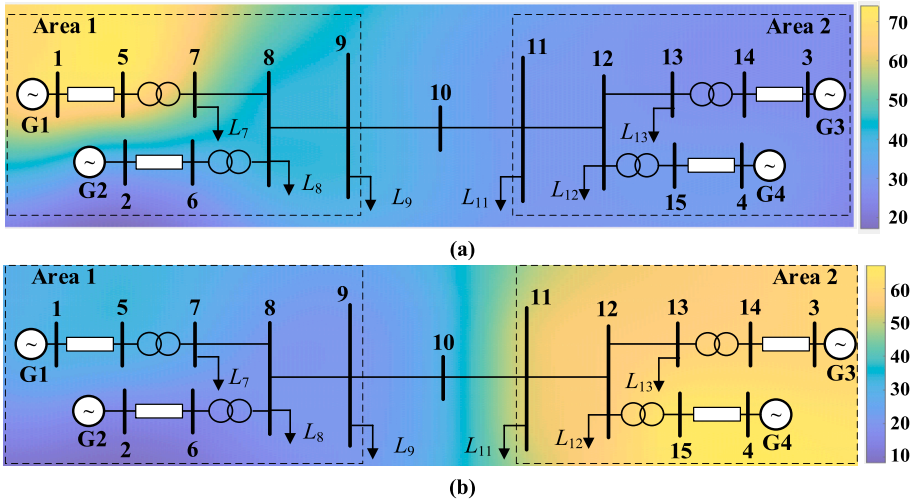


Fig. 8. Inertia distribution with uniform disturbance (α%=15 %) (a) GI in case 1 (b) GI in case 2.

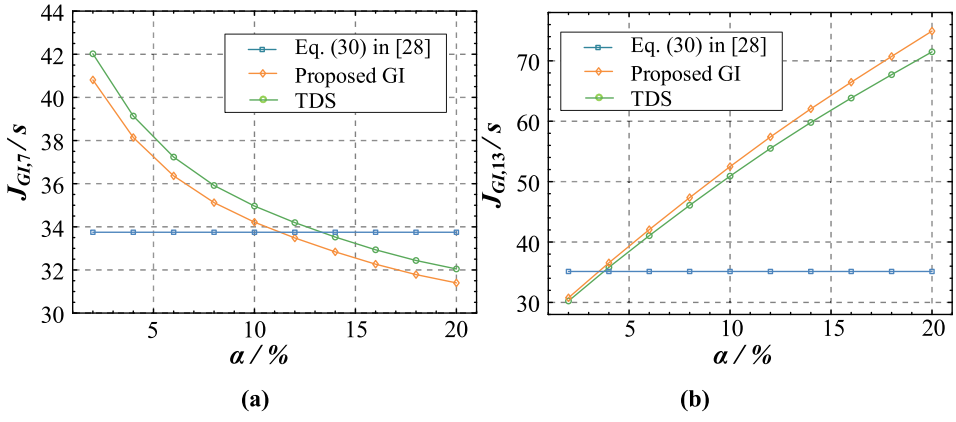


Fig. 9. Inertial value trajectory under non-uniform disturbance (a) Bus 7 (b) Bus 13.

SGs, 4 IBRs and 35 constant impedance loads. The dynamic data of generators is shown in A.ppendix C and the control structure of IBRs is illustrated in A.ppendix B. The control loop parameters of IBRs and the system topology data can be found in [36] and [35] respectively.

The simulation verification of the 5-area 68-bus system was carried out according to the structure of Section 5.1. A power variation of 10 % is applied simultaneously to all load buses and numerical results are shown in Fig. 11 and Fig. 12, which are also observed the effectiveness of the proposed model and the accuracy of the GI method.

The spatial distribution of the inertia is further depicted in Fig. 13.

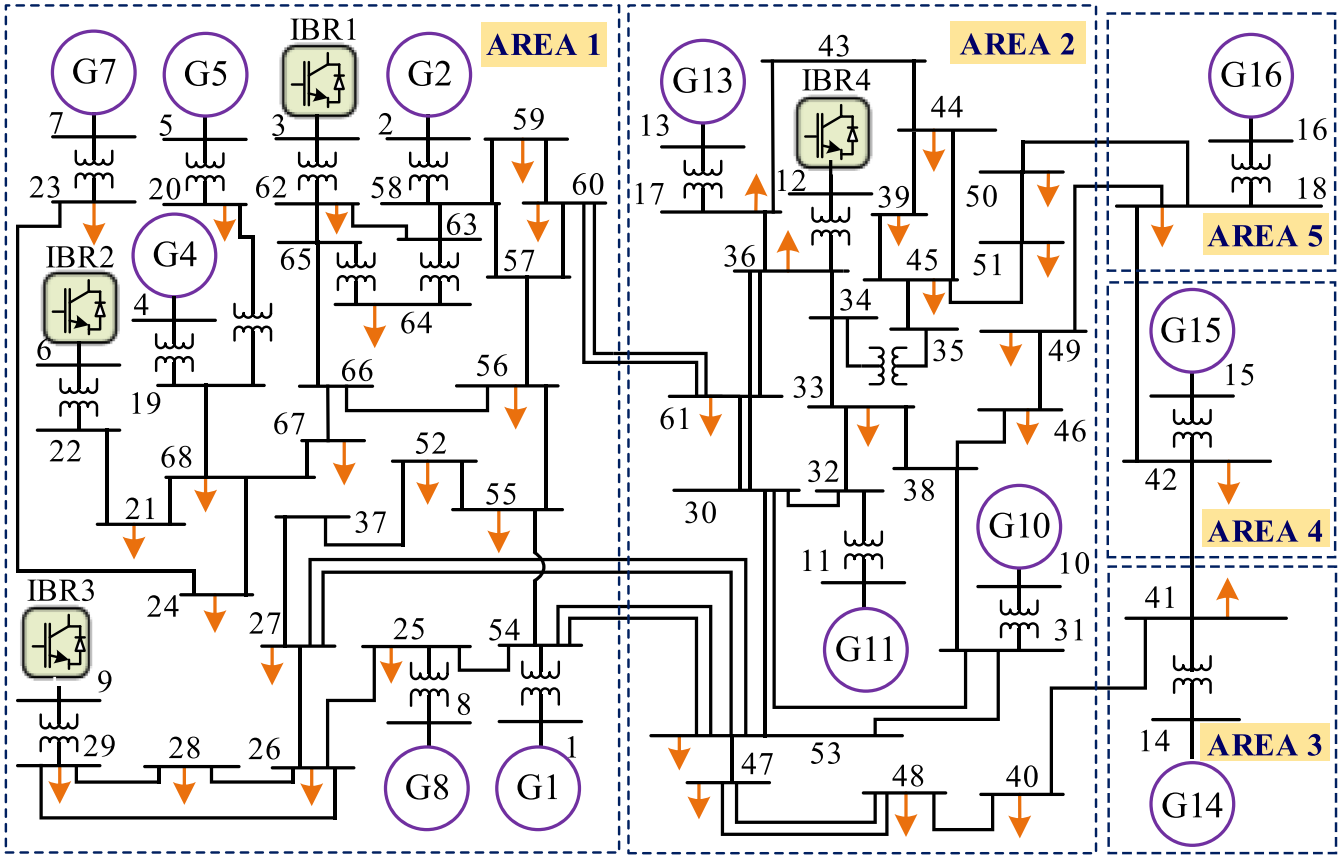


Fig. 10. The 5-area 68-bus system.

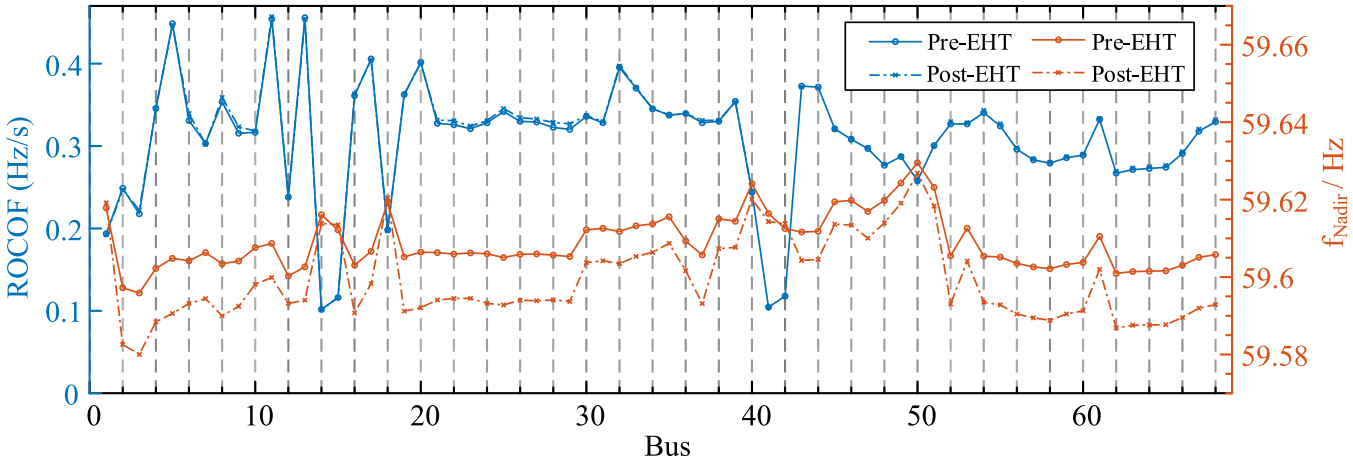


Fig. 11. Frequency dynamic comparison.

The areas near the equivalent generators exhibit high nodal inertia values, peaking at buses 14 and 41 in Area 4 with values of 8,789.2 s and 8,516.8 s, respectively. Low inertia buses are distributed in both Area 1 and Area 2. Bus 5 has the lowest nodal inertia with 1,642.6 s and this can be attributed to the G5′s relatively low inertia constant.

6. Conclusion

This paper proposes an evaluation method of the inertia distribution performance in the multi-machine power system based on the post-EHT FR model. This model meets frequency component-decoupling condition and can reflect the differences among nodal FRs. The nodal inertia is then assessed by the proposed GI-based metric. This metric is an analytical formulation considering both the inertia and disturbance distribution, which can be adopted in the different disturbance scenarios. Finally, the simulation results of the FR trajectory before and after the EHT is well consistent such that the equivalent FR model can reflect the dynamic of a more general system with the generators of heterogeneous parameters. The comparison study of inertia evaluation verifies that the proposed method is better in terms of simulation accuracy than the conventional method, which can only be used in equivalent uniform power change. Given that our approach relies on steady-state parameters and avoids intricate analyses, it can be replicated across various power systems at a small computational expense.

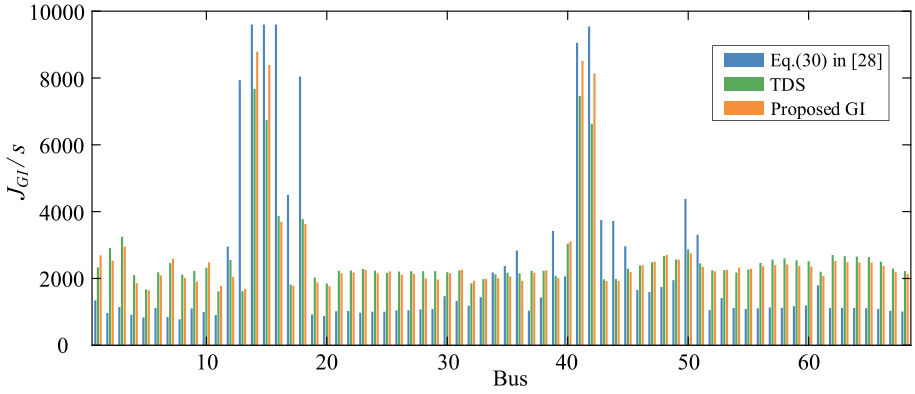


Fig. 12. Inertia comparisons of different methods (α%=10 %).

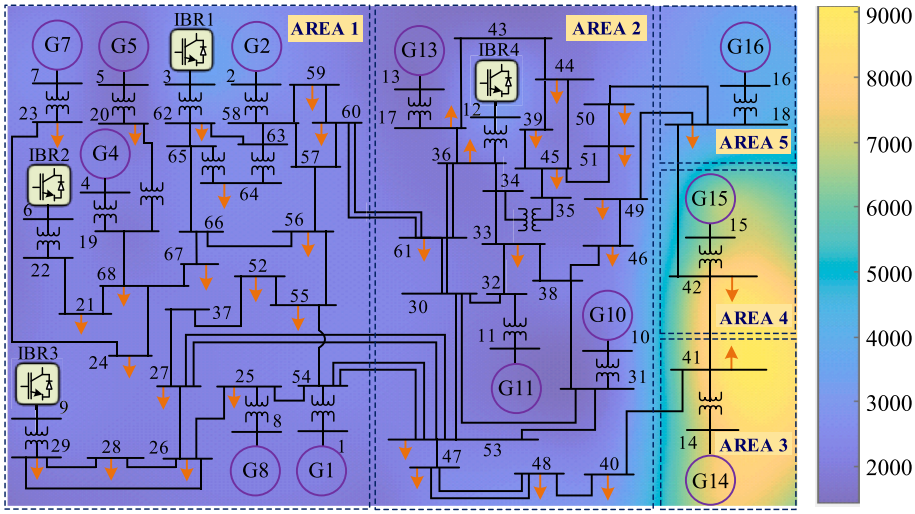


Fig. 13. Inertia distribution for the 5-area 68-bus system.

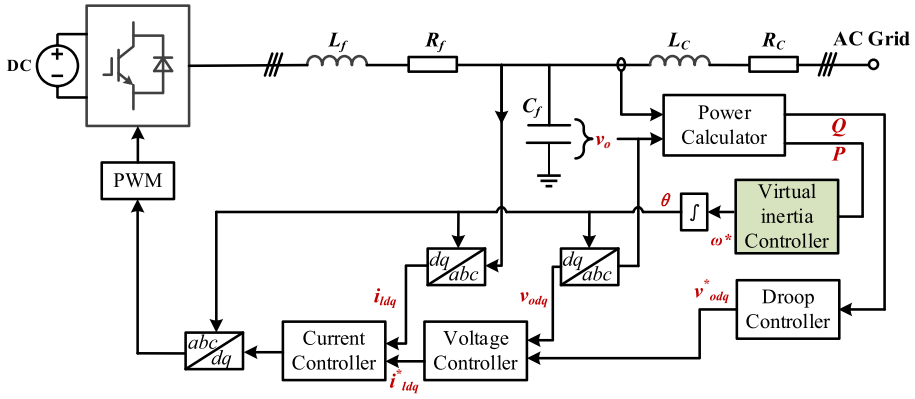


Fig. 14. Grid-Forming IBRs control structure.

CRediT authorship contribution statement

Yu Shan: Methodology, Visualization, Investigation, Writing – original draft. **Zhen Wang:** Conceptualization, Supervision, Formal analysis, Writing – review & editing, Resources, Funding acquisition. **Jialiang Wu:** Supervision, Writing – review & editing. **Huisheng Gao:** Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Frequency dynamics modelling

A.1 Single-machine frequency response model

It is well known that the dynamic relationship between speed and power angle at the *i*-th generator ($i = 1,\ldots,n$) can be represented by [30]:

$$J_i s\Delta\omega_{G,i} + D_i\Delta\omega_{G,i} = \Delta P_{m,i} - \Delta P_{e,i} + \Delta P_{Leq,i} \quad i = 1,\ldots,n \tag{A1}$$

$$s\Delta\theta_{G,i} = \omega_0\Delta\omega_{G,i} \qquad i = 1,\ldots,n \tag{A2}$$

where s is the Laplace operator; J_i and D_i are the inertia constant and damping factor, respectively; $\Delta P_{e,i}$ and $\Delta P_{m,i}$ is the variation of mechanical power and electromagnetic power, respectively; $\Delta P_{Leq,i}$ is equivalent electromagnetic power variation at *i*-th internal buses induced by load disturbance; $\Delta\theta_{G,i}$ is the angle variation of generator internal bus; ω_0 is the reference angular frequency and $\Delta\omega_{G,i}$ is the *i*-th generator's frequency variation; The frequency response model of the single generator can be illustrated in Fig. 2(a).

The primary frequency control can be modelled as a first-order inertial loop, which neglecting the fast dynamics of the governor:

$$T_i s\Delta P_{m,i} + \Delta P_{m,i} = -K_{D,i}\Delta\omega_{G,i} \tag{A3}$$

where T_i is the time constant of turbine control; $K_{D,i}$ is the droop coefficient. From (A1)-(A3), the power angle deviation $\Delta\theta_{G,i}$ and the equivalent electromagnetic power variation $\Delta P_{Leq,i}$ has the following relationship:

$$\Delta\theta_{G,i} = \frac{\omega_0}{s} g_i(s)(\Delta P_{Leq,i} - \Delta P_{e,i}) \tag{A4}$$

where

$$g_i(s) = \frac{T_i s + 1}{J_i T_i s^2 + (D_i T_i + J_i)s + D_i + K_{D,i}} \tag{A5}$$

It is worth noting that that when analyzing the frequency dynamics of the IBRs, the dynamics equation (A5) can also be applicable [37].

A.2 Multi-machine frequency response model

By combining (1) and extending (A4) to a multi-machine system, the frequency response can be written as:

$$\begin{bmatrix} \omega_0^{-1}\mathbf{g}^{-1}(s)s\Delta\boldsymbol{\theta}_G \\ \mathbf{0} \end{bmatrix} = -\begin{bmatrix} \boldsymbol{L}_{11} & \boldsymbol{L}_{12} \\ \boldsymbol{L}_{21} & \boldsymbol{L}_{22} \end{bmatrix}\begin{bmatrix} \Delta\boldsymbol{\theta}_G \\ \Delta\boldsymbol{\theta}_{NI} \end{bmatrix} + \begin{bmatrix} \Delta\boldsymbol{P}_{LG} \\ \Delta\boldsymbol{P}_{LB} \end{bmatrix} \tag{A6}$$

From (A6), the vector $\Delta\boldsymbol{\theta}_{NI}$ can further be:

$$\Delta\boldsymbol{\theta}_{NI} = -\boldsymbol{L}_{22}^{-1}\boldsymbol{L}_{21}\Delta\boldsymbol{\theta}_G + \boldsymbol{L}_{22}^{-1}\Delta\boldsymbol{P}_{LB} \tag{A7}$$

By substituting (A7) into (A6), and it is assumed that there is no instant injected power variation at internal buses when load step suddenly happens, i.e., $\Delta\mathbf{P}_{LG} = \mathbf{0}$. And the system equivalent differential equations can be obtained:

$$\begin{aligned} &\left(\omega_0^{-1}\mathbf{g}^{-1}(s)s + \mathbf{H}_{GG}\right)\Delta\boldsymbol{\theta}_G - \Delta\mathbf{P}_{Leq} = \mathbf{0} \\ &\mathbf{H}_{GG} = \mathbf{L}_{11} - \mathbf{L}_{12}\mathbf{L}_{22}^{-1}\mathbf{L}_{21} \\ &\Delta\mathbf{P}_{Leq} = \mathbf{L}_{12}\mathbf{L}_{22}^{-1}\Delta\mathbf{P}_{\mathrm{LB}} \end{aligned} \tag{A8}$$

The overall block diagram of the frequency response of the multi-machine system is illustrated in Fig. 2(b), which consists of two parts: the network and the generation units.

Considering $\Delta\boldsymbol{\omega}_G = s\Delta\boldsymbol{\theta}_G/\omega_0$, (A8) can be written as

$$\Delta\boldsymbol{\omega}_G = \left(\mathbf{g}^{-1}(s) + \omega_0\mathbf{H}_{GG}/s\right)^{-1}\Delta\mathbf{P}_{Leq} \tag{A9}$$

For further analysis, (A9) is rewritten as

$$\Delta\boldsymbol{\omega}_G = \mathbf{J}^{-1/2}\left(\tilde{\mathbf{g}}^{-1}(s) + \frac{\omega_0}{s}\mathbf{H}_J\right)^{-1}\mathbf{J}^{-1/2}\Delta\mathbf{P}_{Leq} \tag{10}$$

$$\begin{cases} \tilde{\mathbf{g}}^{-1}(s) = diag\left(\tilde{g}_i^{-1}(s)\right) \\ \tilde{g}_i^{-1}(s) = g_i^{-1}(s)/J_i = s + \left(D_i + \dfrac{K_{D,i}}{T_i s + 1}\right)\Big/ J_i \end{cases} \tag{11}$$

A.3 Node frequency response model

Further, by differentiating both sides of (A7), the relationship between the generator frequency deviation $\Delta\boldsymbol{\omega}_G$ and the bus frequency deviation $\Delta\boldsymbol{\omega}_B$ can be derived as:

$$\Delta\boldsymbol{\omega}_B = -\mathbf{L}_{22}^{-1}\mathbf{L}_{21}\Delta\boldsymbol{\omega}_G = \mathbf{D}\Delta\boldsymbol{\omega}_G \tag{12}$$

Here $\Delta\boldsymbol{\omega}_B$ and $\Delta\boldsymbol{\omega}_G$ are the time derivative of $\Delta\boldsymbol{\theta}_{NI}$ and $\Delta\boldsymbol{\theta}_G$, respectively. Substituting (12) into (A12), the node frequency response due to power variation can be derived as:

$$\Delta\boldsymbol{\omega}_B = \mathbf{D}\sum_{l=1}^{n}\mathbf{A}_l\mathbf{A}_l^{\mathrm{T}}G_l(s)\Delta\mathbf{P}_{Leq} \tag{13}$$

Appendix B. The generation units model

B.1 Synchronous generator model

The fifth-order SG model based on [34] considering transient internal voltage is used in the study:

$$\begin{aligned}
&T'se'_d = -e'_d + e_f + (x' - x)i_q \\
&s\frac{\varphi_d}{\omega_0} = v_d - ri_d + \frac{\omega}{\omega_0}\varphi_q \\
&s\frac{\varphi_q}{\omega_0} = v_q - ri_q - \frac{\omega}{\omega_0}\varphi_d \\
&s\theta = \omega \\
&Js\frac{\omega}{\omega_0} = T_e - T_m - D\frac{\omega - \omega_0}{\omega_0}
\end{aligned} \tag{B1}$$

where

$$\begin{aligned}
&\varphi_d = xi_d \\
&\varphi_q = xi_q - e'_d \\
&T_e = e'_d i_d - (x' - x)i_d i_q
\end{aligned} \tag{B2}$$

T'_d and x' are transient time constant and transient internal reactance; e'_d is transient internal voltage; i_d, i_q are d-axis and q-axis stator currents; v_d and v_q are the d-axis and q-axis stator voltage; e_f is the excitation voltage, x and r are stator reactance and resistance, respectively; ω and ω_0 instantaneous and nominal angular speed; θ is the rotor angle, T_e is the electrical torque. The model of the primary frequency control is given in (A3).

B.2 Grid-Forming inverter model

The control structure of Grid-Forming IBRs is shown in Fig. 14, which is based on [36]. The virtual inertia control is formulated as follows

$$s\frac{\omega}{\omega_0} = \left(\Delta P - D_{GFM}\frac{\omega - \omega_0}{\omega_0}\right)\Big/ J_{GFM} \tag{B3}$$

where ΔP is the active power variation; J_{GFM} and D_{GFM} are the virtual inertia and damping factor of GFM, respectively. The structure of voltage controller and current controller can be found in [36].

Appendix C. Dynamic data of 68-Bus system

(See Tables 5-6)

Table 5
SG parameters (p.u., S_N is 100MVA).

SG	x'	X	r	T'_d	J	D	K_D	T
G1	0.0015	0.013	0.002	5.0	84	30.0	150.0	5.0
G2	0.0015	0.035	0.006	5.0	60.4	10.0	70.0	3.5
G4	0.0015	0.030	0.005	5.0	57.2	11.0	90.0	4.2
G5	0.0015	0.027	0.005	5.0	52	13.0	170.0	3.9
G7	0.0015	0.032	0.005	5.0	52.8	9.0	250.0	5.3
G8	0.0015	0.028	0.005	5.0	48.6	20.0	210.0	5.1
G10	0.0015	0.020	0.003	5.0	62	19.0	150.0	7.8
G11	0.0015	0.010	0.002	5.0	56.4	20.0	130.0	4.8
G13	0.0015	0.003	0.001	5.0	496	100.0	210.0	5.1
G14	0.0015	0.002	0.000	5.0	600	160.0	290.0	5.5
G15	0.0015	0.002	0.000	5.0	600	200.0	300.0	3.9
G16	0.0015	0.004	0.001	5.0	600	105.0	250.0	4.6

Table 6
Grid-forming invertor parameters (p.u., S_N is 100MVA).

	Location	J_{GFM}	D_{GFM}
IBR1	Bus 3	71.6	134.9
IBR2	Bus 6	69.6	131.2
IBR3	Bus 9	69.0	130.1
IBR4	Bus 12	184.6	347.9

*all inertia constants for IBRs are from the original SGs.

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